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Intelligent Invoice Processing System

Group 8 – Ema, Redon, Aid

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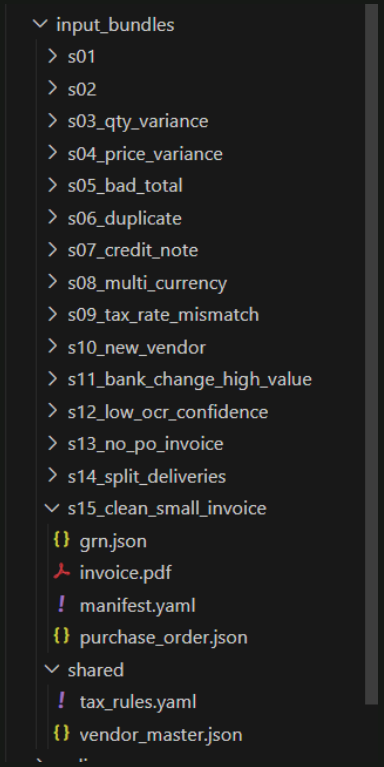
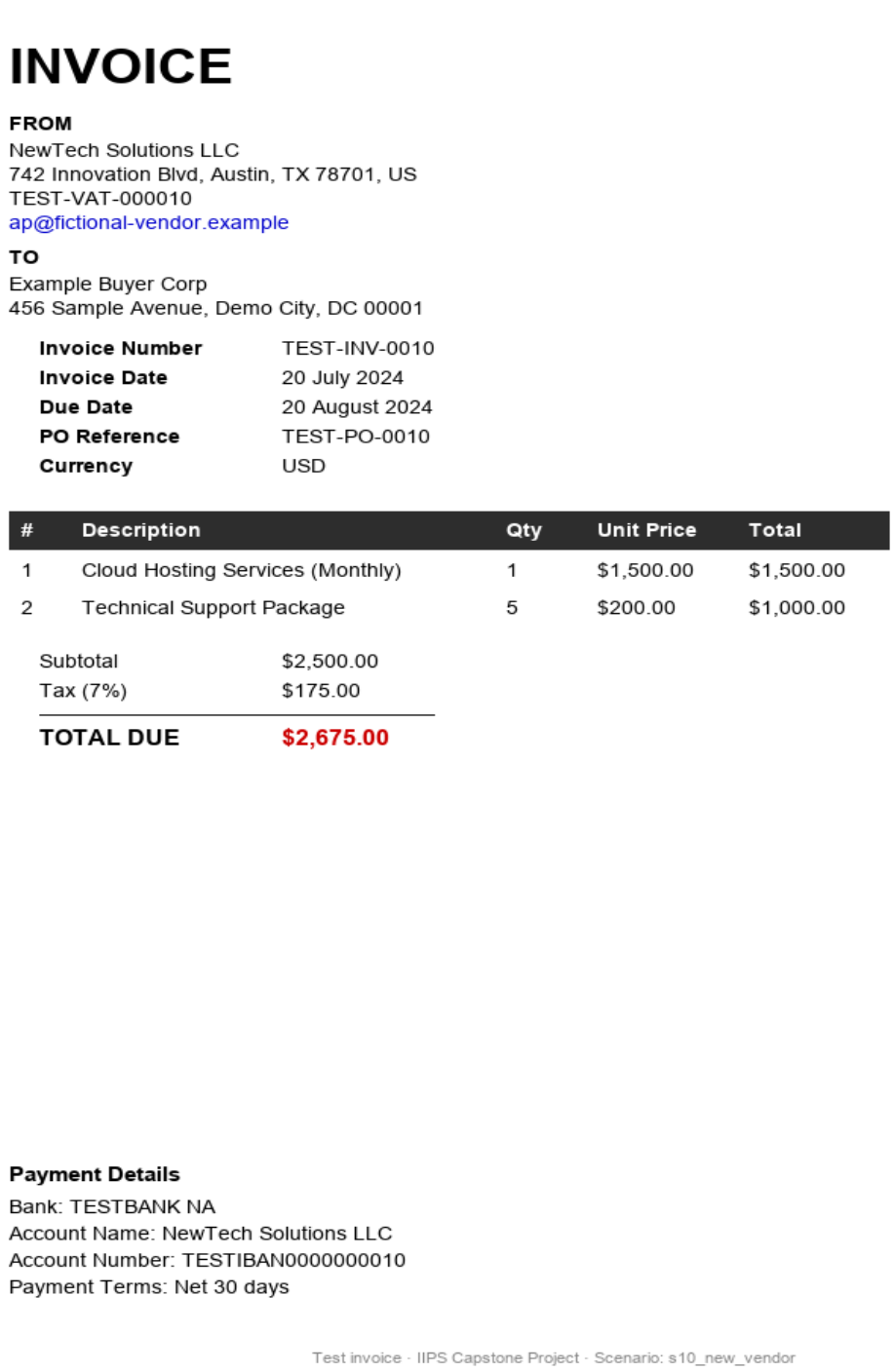
The invoice bundle (input)

- Input: an Invoice Bundle – one folder per invoice
  - Invoice (PDF or PNG)
  - Purchase Order (JSON)
  - Goods Receipt Note (JSON)
  - Vendor Master Data
  - Tax Rules (YAML)
  - Approval Policy (YAML)



- 15 test bundles created – one per scenario (clean invoice, duplicate, new vendor, etc..)



INVOICE

FROM
NewTech Solutions LLC
742 Innovation Blvd, Austin, TX 78701, US
TEST-VAT-000010
ap@fictional-vendor.example

TO
Example Buyer Corp
456 Sample Avenue, Demo City, DC 00001

Invoice Number	TEST-INV-0010
Invoice Date	20 July 2024
Due Date	20 August 2024
PO Reference	TEST-PO-0010
Currency	USD

#	Description	Qty	Unit Price	Total
1	Cloud Hosting Services (Monthly)	1	$1,500.00	$1,500.00
2	Technical Support Package	5	$200.00	$1,000.00

Subtotal	$2,500.00
Tax (7%)	$175.00
TOTAL DUE	**$2,675.00**

Payment Details
Bank: TESTBANK NA
Account Name: NewTech Solutions LLC
Account Number: TESTIBAN0000000010
Payment Terms: Net 30 days

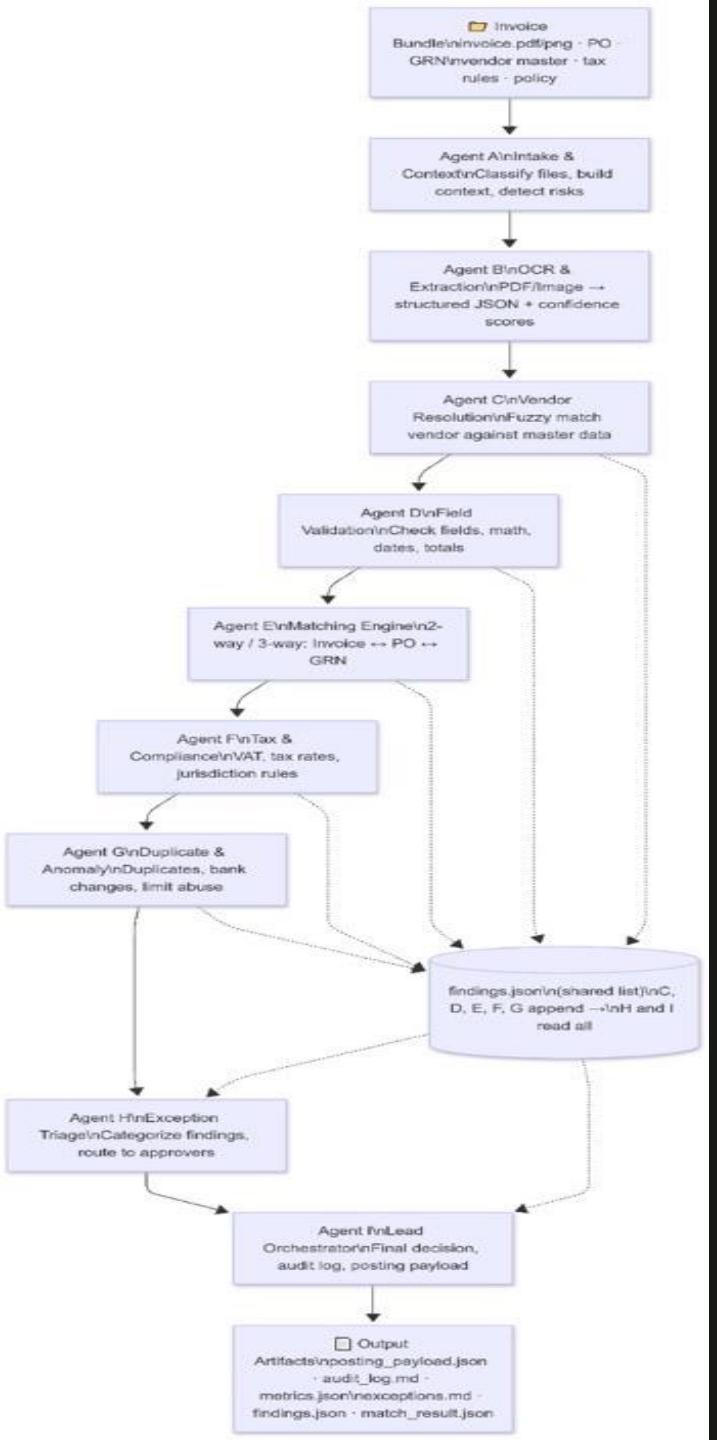
Test invoice · IIPS Capstone Project · Scenario: s10_new_vendor

Pipeline Architecture

9 Specialized agents:
A -> I

Sequential Pipeline -> each agent writes its own output artifact

Full traceability: every decision can be tracked back to its source



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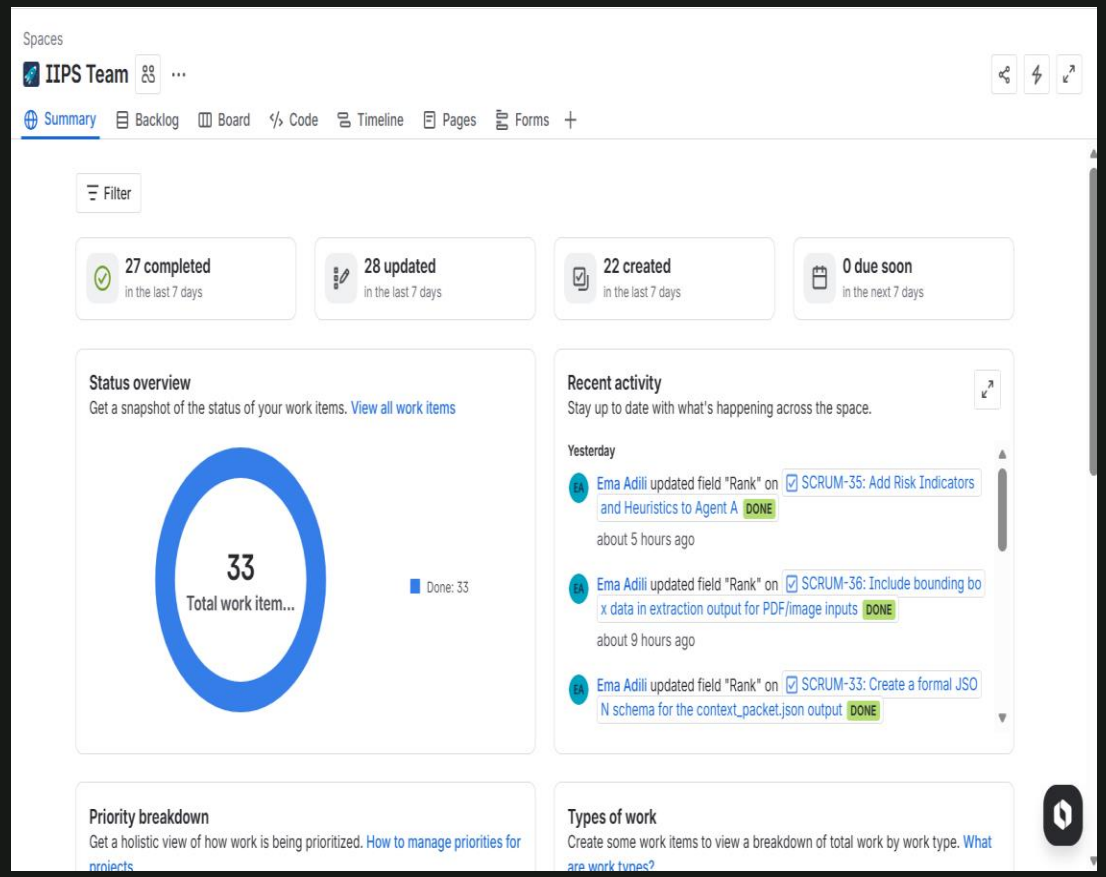
Project managed in Jira – 2 sprints (draft pipeline -> polish & edge cases)



Work	Assignee	Reporter	Priority	Status	Resolution	Created	Updated
SCRUM-20 Build demo.py	Aid Aliu	Aid Aliu	Medium	DONE	Done	Mar 05, 2026, 12:19 PM	Mar 05, 2026, 5:47 F
SCRUM-19 Build run.py	Aid Aliu	Ema Adili	High	DONE	Done	Mar 04, 2026, 12:13 PM	Mar 05, 2026, 5:46 F
SCRUM-18 Agent I — Lead Orchestrator & Final Decision ...	Ema Adili	Ema Adili	High	DONE	Done	Mar 04, 2026, 12:11 PM	Mar 05, 2026, 2:03 F
SCRUM-17 Build agents/agent_h_exception_triage.py → o...	Redon Berisha	Ema Adili	High	DONE	Done	Mar 04, 2026, 12:10 PM	Mar 04, 2026, 8:13 P
SCRUM-15 Agent B uses different function interface than ...	Redon Berisha	Ema Adili	Medium	DONE	Done	Mar 04, 2026, 12:15 AM	Mar 04, 2026, 2:01 P
SCRUM-14 Agent G near-approval-limit check triggers fal...	Redon Berisha	Ema Adili	Medium	DONE	Done	Mar 04, 2026, 12:14 AM	Mar 04, 2026, 12:58
SCRUM-13 Agent F (Compliance) bugs	Ema Adili	Ema Adili	High	DONE	Done	Mar 04, 2026, 12:12 AM	Mar 04, 2026, 2:12 P
SCRUM-12 Create Agent G to detect duplicate invoices an...	Redon Berisha	Ema Adili	High	DONE	Done	Mar 02, 2026, 10:35 PM	Mar 04, 2026, 12:05
SCRUM-11 Create Agent F to validate tax rates against co...	Aid Aliu	Ema Adili	High	DONE	Done	Mar 02, 2026, 10:34 PM	Mar 03, 2026, 11:28 I
SCRUM-10 Create missing test scenario folders for s09 (t...	Ema Adili	Ema Adili	Low	DONE	Done	Mar 02, 2026, 10:33 PM	Mar 03, 2026, 2:33 F
SCRUM-9 Create Agent E that performs 2-way/3-way mat...	Ema Adili	Ema Adili	High	DONE	Done	Mar 02, 2026, 10:31 PM	Mar 03, 2026, 5:46 F
SCRUM-8 Create Agent D to validate extracted invoice da...	Redon Berisha	Ema Adili	High	DONE	Done	Mar 02, 2026, 10:23 PM	Mar 03, 2026, 11:59 I
SCRUM-7 Implement Agent C – Vendor Resolution with F...	Redon Berisha	Ema Adili	Medium	DONE	Done	Mar 02, 2026, 3:12 PM	Mar 02, 2026, 7:22 F
SCRUM-6 Create Configurable Approval Policy YAML (pol...	Redon Berisha	Ema Adili	Medium	DONE	Done	Mar 02, 2026, 3:09 PM	Mar 02, 2026, 4:28 F

Work	Assignee	Reporter	Priority	Status	Resolution	Created	Updated
SCRUM-36 Include bounding box data in extraction o...	Ema Adili	Ema Adili	Medium	DONE	Done	Mar 08, 2026, 10:34 PM	Mar 09, 2026, 2:56 F
SCRUM-35 Include bounding box data in extraction output for PDF/image inputs	Redon Berisha	Ema Adili	Medium	DONE	Done	Mar 08, 2026, 10:33 PM	Mar 09, 2026, 7:26 F
SCRUM-34 Implement universal evidence index generati...	Redon Berisha	Ema Adili	Medium	DONE	Done	Mar 08, 2026, 10:32 PM	Mar 09, 2026, 1:22 P
SCRUM-33 Create a formal JSON schema for the context...	Aid Aliu	Ema Adili	High	DONE	Done	Mar 08, 2026, 10:29 PM	Mar 09, 2026, 2:16 P
SCRUM-32 Create Match Result JSON Schema	Aid Aliu	Ema Adili	High	DONE	Done	Mar 08, 2026, 10:28 PM	Mar 09, 2026, 2:16 P
SCRUM-31 Create a formal JSON schema file for the findi...	Aid Aliu	Ema Adili	High	DONE	Done	Mar 08, 2026, 10:27 PM	Mar 09, 2026, 2:16 P
SCRUM-30 Fix Agent G duplicate detection	Ema Adili	Ema Adili	High	DONE	Done	Mar 08, 2026, 10:06 PM	Mar 09, 2026, 2:05 F
SCRUM-29 Remove unused approval_policy.yaml from in...	Ema Adili	Ema Adili	Low	DONE	Done	Mar 06, 2026, 12:57 PM	Mar 09, 2026, 11:55
SCRUM-28 Agent I's finalize_decision() always uses Age...	Ema Adili	Ema Adili	Medium	DONE	Done	Mar 06, 2026, 12:54 PM	Mar 09, 2026, 11:55
SCRUM-27 Six Scenarios Missing mock_extraction.json (...	Redon Berisha	Ema Adili	Medium	DONE	Done	Mar 06, 2026, 12:52 PM	Mar 09, 2026, 11:55
SCRUM-26 S15 mock extraction has PO reference "PO-00...	Redon Berisha	Ema Adili	Medium	DONE	Done	Mar 06, 2026, 12:51 PM	Mar 09, 2026, 11:55
SCRUM-25 S02 manifest is copy-pasted from S01 and bu...	Redon Berisha	Ema Adili	High	DONE	Done	Mar 06, 2026, 12:46 PM	Mar 09, 2026, 11:43
SCRUM-24 S06 duplicate detection fails on first run -- bu...	Aid Aliu	Ema Adili	High	DONE	Done	Mar 06, 2026, 12:45 PM	Mar 09, 2026, 11:02
SCRUM-23 Duplicate detection causes all scenarios to fai...	Aid Aliu	Ema Adili	High	DONE	Done	Mar 06, 2026, 12:32 PM	Mar 09, 2026, 11:02

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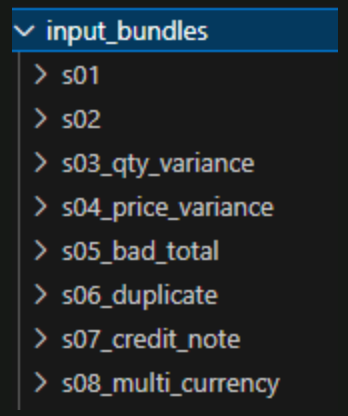
Agent A --- Intake & Evidence Index

Input bundle

* Scenario folder -> one invoice "case"



What Agent A does

* Classifies docs (invoice / PO / GRN / configs)
* Builds **context_packet.json** (metadata + early flags)
* Creates evidence index (source file + path)

Outputs

* **context_packet.json** -> standardized case file
* Evidence pointers -> trace any value to its source
* Findings: **MISSING_PO / NO_GRN** (routing inputs)
* Example: **approval_policy.yaml -> tolerances.quantity_tolerance_pct**

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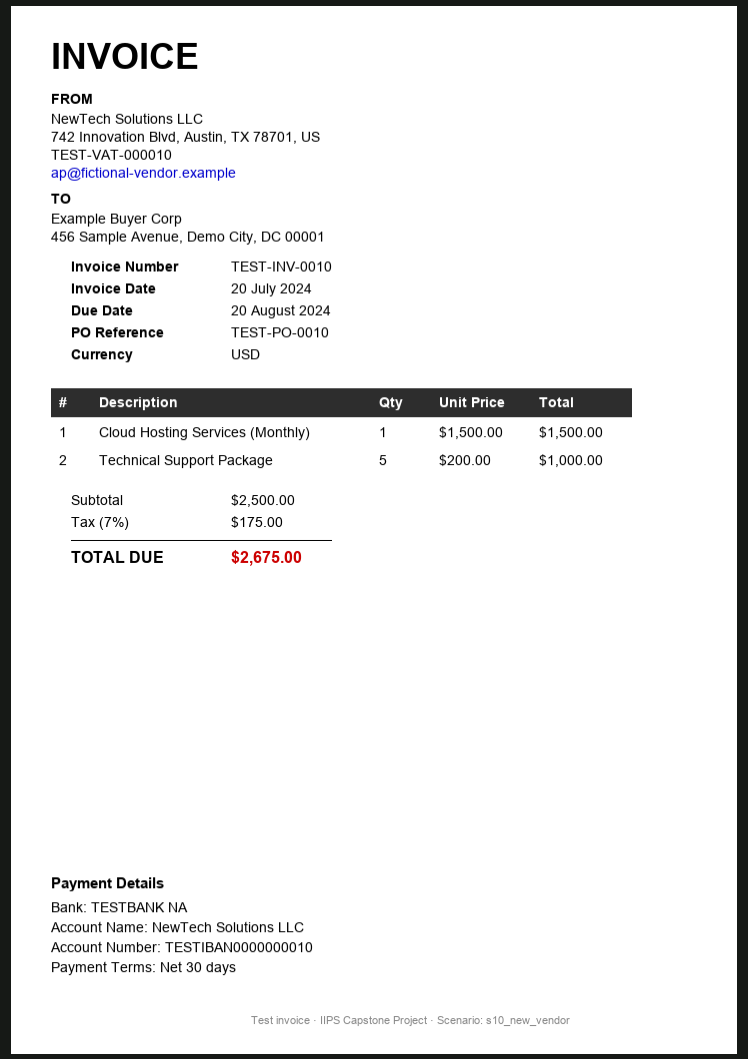
Agent B: OCR & Extraction

- **PDF** : pdfplumber -> GPT-4o-mini -> structured JSON
- **Image (PNG)** : base64 -> OpenAI Vision API -> structured JSON



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Account Number: TESTIBAN0000000010
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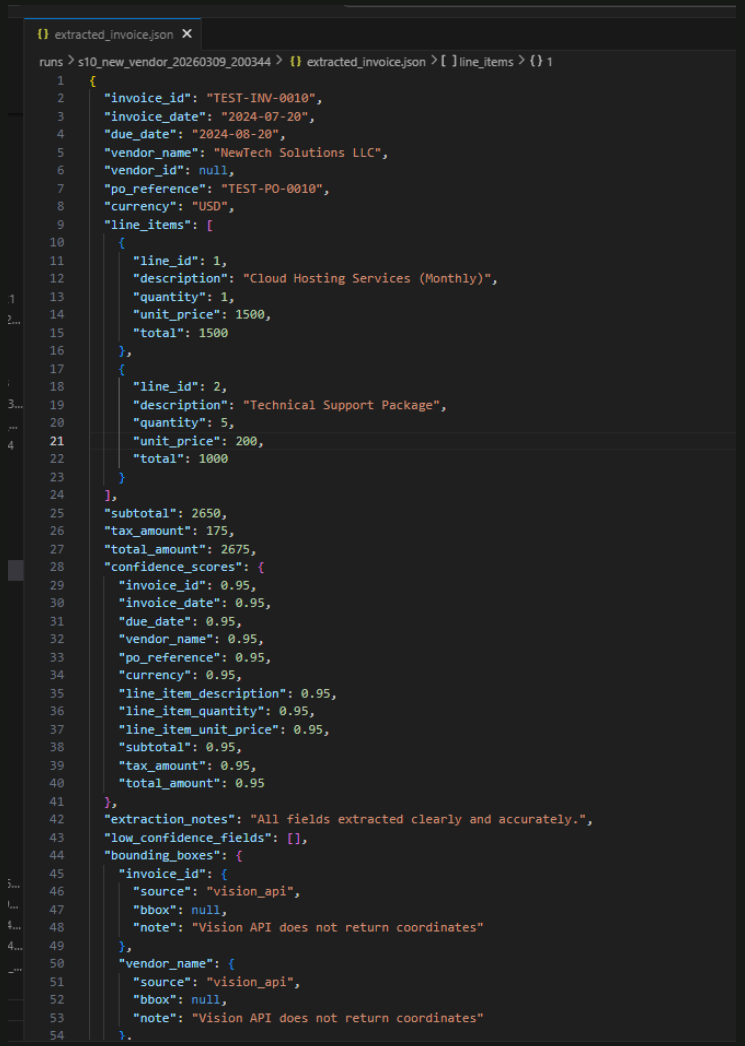
Test invoice · IIPS Capstone Project · Scenario: s10_new_vendor



extracted_invoice.json

runs > s10_new_vendor_20260309_200344 > extracted_invoice.json > [ ] line_items > { } 1


```
{
  "invoice_id": "TEST-INV-0010",
  "invoice_date": "2024-07-20",
  "due_date": "2024-08-20",
  "vendor_name": "NewTech Solutions LLC",
  "vendor_id": null,
  "po_reference": "TEST-PO-0010",
  "currency": "USD",
  "line_items": [
    {
      "line_id": 1,
      "description": "Cloud Hosting Services (Monthly)",
      "quantity": 1,
      "unit_price": 1500,
      "total": 1500
    },
    {
      "line_id": 2,
      "description": "Technical Support Package",
      "quantity": 5,
      "unit_price": 200,
      "total": 1000
    }
  ],
  "subtotal": 2650,
  "tax_amount": 175,
  "total_amount": 2675,
  "confidence_scores": {
    "invoice_id": 0.95,
    "invoice_date": 0.95,
    "due_date": 0.95,
    "vendor_name": 0.95,
    "po_reference": 0.95,
    "currency": 0.95,
    "line_item_description": 0.95,
    "line_item_quantity": 0.95,
    "line_item_unit_price": 0.95,
    "subtotal": 0.95,
    "tax_amount": 0.95,
    "total_amount": 0.95
  },
  "extraction_notes": "All fields extracted clearly and accurately.",
  "low_confidence_fields": [],
  "bounding_boxes": {
    "invoice_id": {
      "source": "vision_api",
      "bbox": null,
      "note": "Vision API does not return coordinates"
    },
    "vendor_name": {
      "source": "vision_api",
      "bbox": null,
      "note": "Vision API does not return coordinates"
    },
```


- **Outputs**: extracted_invoice.json & line_items.csv
- Bounding boxes per field

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Agent C – Vendor Resolution

How it works, step by step:

Its job is to answer: "Which approved vendor in our system does this invoice actually belong to"

- Reads the extracted invoice from agent B and pulls out the vendor name text.
- Loads the vendor master – the approved list of vendors with their IDs, names, and flags.
- Fuzzy-matches the name using a library called "rapidfuzz". It compares the invoice's vendor name against every vendor in the master and scores each one 0.0 - 1.0 based on how similar the names are. 1 being perfect match.
- Sorts by score, then alphabetically by vendor_id as a tiebreaker. It picks the top result as the winner.
- Checks against a confidence threshold (default 0.75 from the policy). If the best match scores below that, its flagged as a WEAK_MATCH risk. If nothing matches at all, it's NEW_VENDOR.
- Writes vendor_resolution_result.json with the matched vendor_id, confidence score, status, and the top 3 candidates.

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Agent D – Invoice validation

Its job is to answer: "Is this invoice internally consistent and structurally correct?

It runs 9 validations checks

1: Required fields – Missing ID, date, vendor, amounts

2: Date validity – Future dates, due date before invoice date

3: Currency code – Invalid format (must be 3-letter ISO e.g. USD)

4: Line item Math – qty x price = line total

5: Subtotal – Adds up all line item totals and verifies they match invoice's subtotal

6: Header Total – Verifies that subtotal + tax_amount = total_amount

7: PO Reference – Invoice over $1,000 with no PO number

8: Credit Note – Negative total flagged for special handling

9: OCR Confidence – Low-confidence fields flagged for manual review

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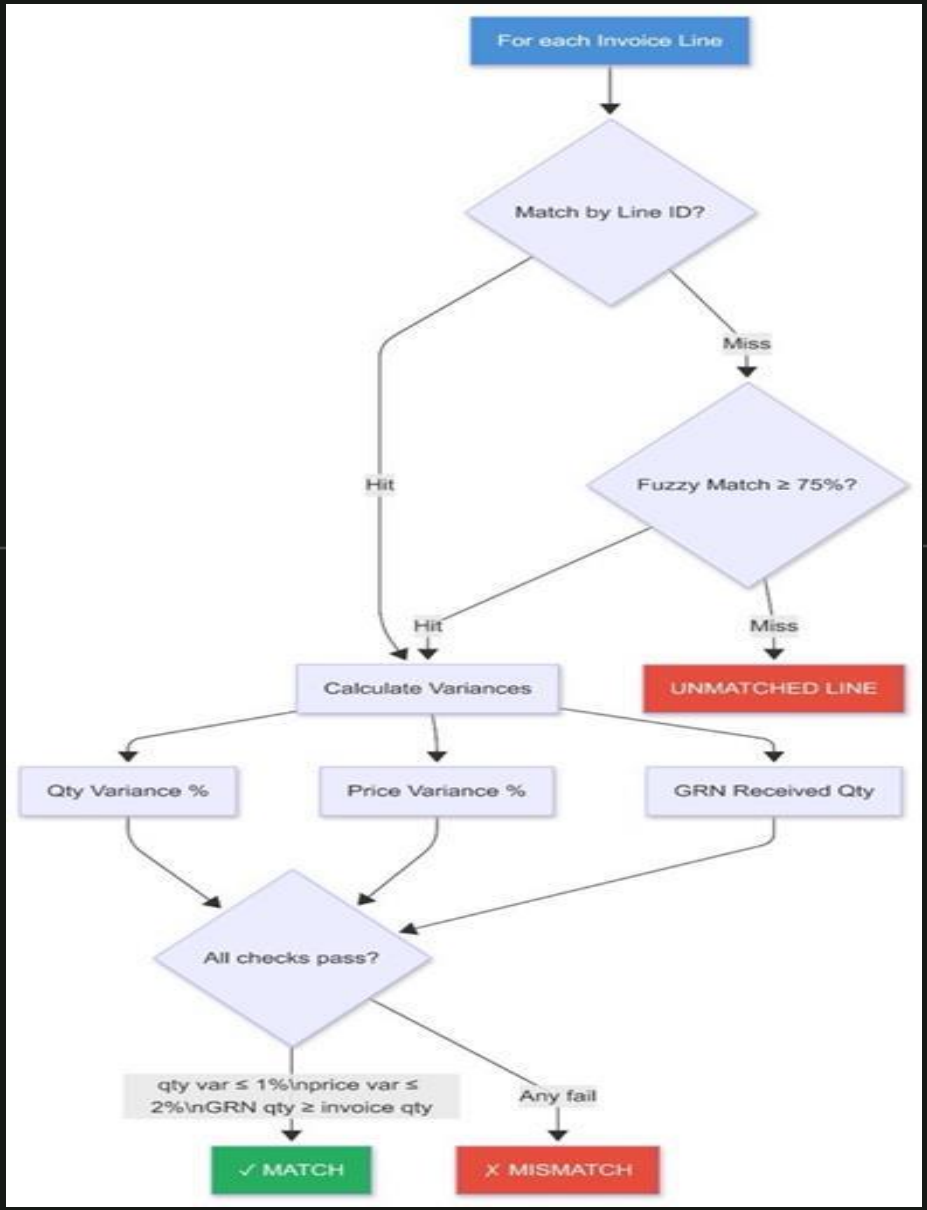
Agent E: Matching Engine

2 way matching (Invoice vs PO)

Or 3 way matching (Invoice vs PO vs GRN)

Split deriveries: aggregates GRN quantities per PO line



Line matching: by ID first, fallback fuzzy matching (RapidFuzz, 75%)

Variance checks against YAML tolerances (price: 2%, quantity: 1%)

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Agent F: Tax Compliance

What it checks

- ✓ VAT/tax rate vs tax_rules.yaml
- ✓ Tax amount math (subtotal × rate)
- ✓ VAT ID format sanity check

→

Extracted totals (subtotal, tax)

Expected rate (tax_rules.yaml)

Compare + tolerance

↓

Compare actual tax rate with standard/reduced country rate

→

Outputs:

- **Findings:**
  - TAX_RATE_MISMATCH
  - TAX_AMOUNT_MISMATCH
- **compliance_result.json** (PASS/FAIL + details)
- **Evidence pointer** → rule used (country/vendor)

```
actual_rate = tax_amount / subtotal

rules_list = tax_rules.get("tax_rules", [])
rule = next((r for r in rules_list if r['country_code')

if rule:
    expected_rate = float(rule.get('standard_rate', 0))
    reduced_rate = float(rule.get('reduced_rate', 0))
    tolerance = float(policy.get('tax_rules', {}).get('ta

    diff_std = abs(actual_rate - expected_rate)
    diff_red = abs(actual_rate - reduced_rate)

    match_std = diff_std <= tolerance
    match_red = diff_red <= tolerance
```

+ tax_rules.yaml

input_bundles > shared > **tax_rules.yaml**

```yaml
tax_rules:
  - country_code: "US"
    standard_rate: 0.07  # 7% standard sales tax
    reduced_rate: 0.00  # 0% for exempt items
  - country_code: "GB"
    standard_rate: 0.20  # 20% standard VAT
    reduced_rate: 0.08  # 5% reduced VAT (e.g, uilities
  - country_code: "DE"
    standard_rate: 0.19  # 19% standard VAT
    reduced_rate: 0.07  # 7% reduced VAT
```

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Agent G – Anomaly Detection

Duplicate Invoice

It triggers when: Same vendor, invoice number, date & amount seen before => DUPLICATE_INVOICE - BLOCK

Bank Change (High Value)

It triggers when: Vendor's bank details recently changed + total > $5,000 =>

BANK_CHANGE_HIGH_VALUE – Escalate to Finance

Near Approval Limit

It triggers when: Invoice total is within top 2% of the senior approval threshold =>

NEAR_APPROVAL_LIMIT -

Manual review

"Does anything about this invoice look suspicious or fraudulent"

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Agent H – Exception Triage

"Given everything they found, what do we do with this invoice"

Collects all findings from agetns C – G and applies routing rules in priority order:

Condition		Action
1. Duplicate invoice detected	->	BLOCK -> Risk Officer
2. Uknown or unresolved vendor	->	HOLD_FOR_APPROVAL -> AP Manager
3. Low OCR confidence	->	HOLD_FOR_MANUAL_REVIEW -> AP Manager
4. Any Critical findings	->	ESCALATE_TO_RISK_OFFICER
5. Total > senior threshold	->	HOLD_FOR_APPROVAL -> Finance Controller
6. Any findings at all	->	HOLD_FOR APPROVAL -> AP Manager
7. Total > auto-approve limit	->	HOLD_FOR APPROVAL -> AP Manager
8. Clean + umder limit	->	AUTO_APPROVE -> AP Bot

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Agent I: Lead Orchestrator

- Collects all output → deduplicates & sorts findings
- Decision logic:
  - Clean invoice → auto-post
  - Multi-currency → route for approval
  - Vendor issues / missing PO → department head
  - Bank account change (high-value) → finance controller
  - Duplicate → blocked
- Final artifacts:
  - Posting_payload.json → ERP-ready decision
  - Audit_kog.md → full trace across all 9 agents
  - Metrics.json → confidence scores + determinism

Preview audit_log.md

Audit Log — Invoice TEST-INV-0010

Generated: 2026-03-09 20:03:57 **Determinism Hash:** `02ca05a639eaeb8def11707616769c0c47ae885eec61c6874accb70fbbaddf45`

Run Metadata

Field	Value
Invoice ID	TEST-INV-0010
Vendor	NewTech Solutions LLC (None)
Total Amount	USD 2,675.00
PO Reference	TEST-PO-0010
Final Action	**ROUTE_TO_DEPT_HEAD**
Assigned To	dept_head
Hold Reasons	Vendor resolution required — routing to department head

Agent Trace

Step	Agent	Context Key	Status	Detail
1	A — Intake	run_id / status	completed	run_id=s10_new_vendor_20260309_200344
2	B — Extraction	files	completed	6 files
3	C — Vendor Resolution	vendor_resolution	WEAK_MATCH	confidence=0.4737
4	D — Field Validation	field_validation	2 findings	passed=False
5	E — Matching	matching	FULL_MATCH	match_type=3WAY
6	F — Tax Compliance	tax_validation	PASS	jurisdiction=N/A
7	G — Anomaly Detection	anomaly_detection	CLEAR	—
8	H — Exception Triage	exception_triage	HOLD_FOR_APPROVAL	assigned_to=ap_manager
9	I — Orchestrator	orchestrator	ROUTE_TO_DEPT_HEAD	assigned_to=dept_head

Final Decision

Field	Value
Action	ROUTE_TO_DEPT_HEAD
Assigned To	dept_head
Approval Required	Yes
Posting Blocked	No
Determinism Hash	`02ca05a639eaeb8def11707616769c0c47ae885eec61c6874accb70fbbaddf45`

Findings (3)

#	Agent	Code	Severity	Message
1	C	VENDOR_MATCH_WEAK	MEDIUM	Vendor confidence below threshold.
2	D	HEADER_TOTAL_MISMATCH	HIGH	subtotal(2650) + tax(175) = 2825, but total_amount = 2675.
3	D	SUBTOTAL_MISMATCH	HIGH	sum(line_items.total) = 2500 but subtotal = 2650.

Evidence File Paths

- **anomaly_detection_result.json:** `C:\Users\602000848\Desktop\ai-invoice-processor\runs\s10_new_vendor_20260309_200344\anomaly_detection_result.json`
- **approval_packet.json:** `C:\Users\602000848\Desktop\ai-invoice-processor\runs\s10_new_vendor_20260309_200344\approval_packet.json`
- **context_packet.json:** `C:\Users\602000848\Desktop\ai-invoice-processor\runs\s10_new_vendor_20260309_200344\context_packet.json`
- **exceptions.md:** `C:\Users\602000848\Desktop\ai-invoice-processor\runs\s10_new_vendor_20260309_200344\exceptions.md`
- **extracted_invoice.json:** `C:\Users\602000848\Desktop\ai-invoice-processor\runs\s10_new_vendor_20260309_200344\extracted_invoice.json`
- **findings.json:** `C:\Users\602000848\Desktop\ai-invoice-processor\runs\s10_new_vendor_20260309_200344\findings.json`
- **match_result.json:** `C:\Users\602000848\Desktop\ai-invoice-processor\runs\s10_new_vendor_20260309_200344\match_result.json`
- **tax_validation_result.json:** `C:\Users\602000848\Desktop\ai-invoice-processor\runs\s10_new_vendor_20260309_200344\tax_validation_result.json`
- **validation_result.json:** `C:\Users\602000848\Desktop\ai-invoice-processor\runs\s10_new_vendor_20260309_200344\validation_result.json`
- **vendor_resolution_result.json:** `C:\Users\602000848\Desktop\ai-invoice-processor\runs\s10_new_vendor_20260309_200344\vendor_resolution_result.json`

Pipeline Execution – run.py

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